

CO2 - AT2 - Scenario Based Questions

CSA0203 – C Programming

Q1 Scenario Based - 1. Electric Vehicle Battery Manufacturing System

25 marks

An electric vehicle battery manufacturing company produced batteries from January 2026 to December 2026. Each month, the target production is 5,000 batteries. If production exceeds the target, the company receives a 10% performance incentive. If production is below 80% of the target, the production unit is marked for inspection.

A) Develop a C program using arithmetic operators to calculate the annual production, average monthly production, and production efficiency. Use decision-making statements to determine the incentive and inspection status.

B) Use looping constructs to process production details for all 12 months. Use continue to skip months with invalid production values and break if the plant shuts down due to equipment failure.

Your Code

```
1 #include <stdio.h>
2
3 void main()
4 {
5     int production[12];
6     int i;
7     int annualProduction = 0;
8     float averageProduction, efficiency;
9
10    printf("Enter production for 12 months:\n");
11
12    for (i = 0; i < 12; i++) {
13        scanf("%d", &production[i]);
14
15
16        if (production[i] < 0) {
17            continue;
18        }
19
20
21        if (production[i] == -1) {
22            break;
23        }
24
25        annualProduction += production[i];
26
27
28        if (production[i] > 5000) {
29            printf("Month %d: 10%% Performance Incentive\n", i + 1);
30        }
31        else if (production[i] < 4000) {
32            printf("Month %d: Production Unit Marked for Inspection\n", i + 1);
33        }
34        else {
35            printf("Month %d: Normal Production\n", i + 1);
36        }
37    }
38
39    averageProduction = annualProduction / 12.0;
40    efficiency = (annualProduction / 60000.0) * 100;
41
42    printf("\nAnnual Production = %d batteries\n", annualProduction);
43    printf("Average Monthly Production = %.2f batteries\n", averageProduction);
44    printf("Production Efficiency = %.2f%%\n", efficiency);
45 }
```

Input

5200
4500
3800
5100
...

Output

Enter production for 12 months:
Month 1: 10% Performance
Incentive
Month 2: Normal Production
Month 3: Production Unit Marked
for Inspection

A power distribution company monitors 15 substations every hour. Each substation has a maximum capacity of 500 MW.

Business Rules:

Load $\leq$ 60% → Normal

61–85% → High Load

Above 85% → Critical

Above 500 MW → Grid Failure

A) Develop a C program using arithmetic operators to calculate the load percentage and available capacity. Use decision-making statements to classify each substation.

B) Use looping constructs to monitor all substations. Use continue to skip substations under maintenance and break if grid failure occurs.

Your Code

```
1 #include <stdio.h>
2
3 void main()
4 {
5     int i;
6     float load, percentage, available;
7     int status;
8
9     for (i = 1; i <= 15; i++)
10    {
11        printf("Enter load for Substation %d (MW): ", i);
12        scanf("%f", &load);
13
14        if (load < 0)
15        {
16            printf("Substation %d: Under Maintenance\n", i);
17            continue;
18        }
19
20
21        if (load > 500)
22        {
23            printf("Substation %d: Grid Failure\n", i);
24            break;
25        }
26
27        percentage = (load / 500) * 100;
28
29        available = 500 - load;
30
31
32        if (percentage <= 60)
33        {
34            printf("Substation %d: Normal\n", i);
35        }
36        else if (percentage <= 85)
37        {
38            printf("Substation %d: High Load\n", i);
39        }
40        else
41        {
42            printf("Substation %d: Critical\n", i);
43        }
44
45        printf("Load Percentage = %.2f%%\n", percentage);
```

Input

250
350
450
550

Output

Enter load for Substation 1
(MW): Substation 1: Normal
Load Percentage = 50.00%
Available Capacity = 250.00 MW
Enter load for Substation 2

Q3 Scenario Based - 3. A pharmaceutical company manufactured 2,50,000 vaccine doses in 2026

25 marks

Vaccines are distributed to 25 hospitals. Each hospital must receive at least 8,000 doses.

Distribution Status:

≥10,000 doses → Sufficient

8,000–9,999 → Minimum Requirement

Below 8,000 → Emergency Supply Required

A) Develop a C program using arithmetic operators to calculate remaining vaccine stock after distribution. Use decision-making statements to classify each hospital.

B) Use looping constructs to process all 25 hospitals. Use continue for closed hospitals and break if stock becomes zero.

Your Code

```
1 #include <stdio.h>
2
3 void main()
4 {
5     int i, doses;
6     int stock = 250000;
7
8     for (i = 1; i <= 25; i++)
9     {
10         printf("Enter doses for Hospital %d: ", i);
11         scanf("%d", &doses);
12
13         if (doses == -1)
14         {
15             printf("Hospital %d: Closed\n", i);
16             continue;
17         }
18
19         if (stock == 0)
20         {
21             printf("Stock is zero. Distribution stopped.\n");
22             break;
23         }
24
25         stock = stock - doses;
26
27         if (doses >= 10000)
28         {
29             printf("Hospital %d: Sufficient\n", i);
30         }
31         else if (doses >= 8000)
32         {
33             printf("Hospital %d: Minimum Requirement\n", i);
34         }
35         else
36         {
37             printf("Hospital %d: Emergency Supply Required\n", i);
38         }
39
40         printf("Remaining Stock = %d doses\n\n", stock);
41     }
42
43     printf("Final Remaining Vaccine Stock = %d doses\n", stock);
44 }
```

Input

10000
9000
7500
12000

Output

Enter doses for Hospital 1:
Hospital 1: Sufficient
Remaining Stock = 240000 doses

Enter doses for Hospital 2:
Hospital 2: Minimum Requirement

An international airport processes 1,500 passengers daily.

Security Rules:

Age below 18 → Minor Screening

Age 18–59 → Normal Screening

Age 60 and above → Senior Citizen Screening

Carrying baggage above 30 kg → Additional Inspection

Carrying prohibited items → Entry Denied

A) Develop a C program using arithmetic operators to calculate total baggage weight and screening statistics. Use decision-making statements to determine passenger screening status.

B) Use looping constructs to process passengers. Use continue to skip cancelled tickets and break during emergency evacuation.

Your Code

```
1 #include <stdio.h>
2
3 void main()
4 {
5     int i, age;
6     float baggage, totalBaggage = 0;
7     int prohibited;
8
9     for (i = 1; i <= 1500; i++)
10    {
11        printf("Enter age, baggage weight and prohibited item (1-Yes, 0-No)\n");
12        scanf("%d %f %d", &age, &baggage, &prohibited);
13
14        if (age < 0)
15        {
16            printf("Passenger %d: Cancelled Ticket\n", i);
17            continue;
18        }
19
20        if (age == 0)
21        {
22            printf("Emergency Evacuation!\n");
23            break;
24        }
25
26        totalBaggage = totalBaggage + baggage;
27
28        if (age < 18)
29        {
30            printf("Passenger %d: Minor Screening\n", i);
31        }
32        else if (age <= 59)
33        {
34            printf("Passenger %d: Normal Screening\n", i);
35        }
36        else
37        {
38            printf("Passenger %d: Senior Citizen Screening\n", i);
39        }
40
41        if (baggage > 30)
42        {
43            printf("Additional Inspection Required\n");
44        }
45    }
```

Input

```
20 25 0
16 15 0
65 35 0
30 20 1
```

Output

```
Enter age, baggage weight and
prohibited item (1-Yes, 0-No)
for Passenger 5: Emergency
Evacuation!
Total Baggage Weight = 95.00 kg
```